

VISWESWAR SIRISH PARUPUDI

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EDUCATION

New York University

Master of Science, Computer Science | GPA: 3.85

Brooklyn, NY

May 2026

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering, Electronics and Communication | Minor in Computing and Intelligence | GPA: 3.45

Hyd, India

May 2024

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, C#, MATLAB, HTML/CSS

Frameworks: React, Django, Express/Node.js, REST APIs, Docker, Kubernetes, Kafka, Spark, Redis, Git, CI/CD

ML/Cloud & Tools: AWS (EC2, Elastic Beanstalk, RDS, S3), PostgreSQL, PyTorch, TensorFlow, Keras, NumPy, Pandas, OpenAI Gym, Unity

EXPERIENCE

Research Assistant

June 2025 – Present

Neuroinformatics Lab, New York University

New York, NY

- Built a framework for real-time in-vivo brain region localization from raw multichannel LFP on **5 TB** of mouse neural recordings spanning the full brain atlas (**307-class** localization task), eliminating dependence on post-hoc histology (submission for **NeurIPS 2026**).
- Introduced novel geometry aware pretraining objectives; achieving state-of-the-art coordinate regression; 1.65 mm error across the 1 cm mouse brain, with zero-shot generalization across probe types, laboratories, and species.
- Maintain and develop the lab's public-facing website, managing deployment, version control, and content updates via Git to accurately represent ongoing research and publications.

Graduate Assistant | Course Links: Computer Vision, Neuroinformatics

Aug. 2025 – Present

New York University

New York, NY

- Facilitated graduate-level live coding sessions and office hours for **100+ students**; provided mentorship and guidance for projects.
- Authored and graded assignments/exams and curated multiple Kaggle-style competitions (dataset curation, baselines, and metrics); evaluated submissions and delivered structured feedback across core and advanced CV and Neuroscience topics.

Software Engineer Intern

May 2022 – July 2022

DLT Labs (KNNX)

Hyderabad, India

- Designed a real-time forecasting service for Ethereum gas prices, integrating with **Ethereum JSON-RPC APIs** for continuous data ingestion and preprocessing at minute-level granularity; provisioned downstream-ready daily and real-time prediction endpoints.
- Reduced **prediction error by 27%** through systematic ARIMA model selection, ACF/PACF diagnostics, and grid-search hyperparameter tuning; accurate forecasts enabled downstream systems to optimize transaction timing, reducing gas fee expenditure for end users.
- Implemented data quality checks and event-driven alerting for pipeline failures, ensuring **99%+ uptime** of the forecasting service; wrote unit and integration tests to validate model outputs against expected price ranges and API response schemas.

PROJECTS

CineMatch – Group Movie Matching & Social Discovery Platform | React, Django, PostgreSQL, AWS | [Link](#)

- Built a full-stack collaborative movie discovery platform with a normalized relational schema, RESTful microservices, and **WebSocket-backed** real-time chat with Redis pub/sub for low-latency group matching and synchronized swiping.
- Integrated an **Agentic recommendation engine** with personalized scoring, genre playlists, group spaces with visibility controls, and watchlist import; implemented full-text search across a 50K+ title catalog via external API integration.
- Led** a 6-person agile team across 8 sprints; established **CI/CD pipelines** with 90%+ test coverage and deployed to production on AWS EC2/Elastic Beanstalk, achieving sub-300ms chat delivery under concurrent load.

On-Device Myocardial Infarction Detection via ECG | Python, TFLite, Android | Publication: IEEE Sensors Letters (2023)

- Optimized a cardiac detection model for edge deployment via FP16/INT8 quantization, achieving $\sim 2.3\times$ **compression** under strict mobile memory and latency constraints; deployed end-to-end on Android via camera-captured ECG trace classification.
- Designed a majority-voting ensemble with under 32K parameters for low-cost IoT-based cardiac monitoring; **published in IEEE Sensors Letters** with 99.34% accuracy via five-fold CV, outperforming all individual baselines.

Text Augmentation for NER on Noisy Language (Undergraduate Thesis) | Python, PyTorch, LangChain | [Thesis Link](#)

- Developed an end-to-end augmentation pipeline to convert noisy tweets into clean training data for Named Entity Recognition (NER) without re-pretraining, using entity placeholder substitution to prevent label drift during paraphrase generation.
- Architected a multi-sample paraphrase generation system with BERTScore-guided candidate selection for entity retention, and a fuzzy-matching retagging module to re-align clean-domain predictions back to original noisy tokens for ground truth comparison.
- Extended pipeline with LLaMA 2 few-shot prompting to replace PEGASUS-based paraphrasing, improving overall NER accuracy by $\sim 8.6\%$ over baseline with only 6% entity loss and **24% fluency gain** on augmented text.

Scalable Chest X-Ray Triage Pipeline | Kafka, Spark, AWS S3, Python, Docker | [Link](#)

- Engineered a scalable, production-style pipeline for automated chest X-ray triage, supporting end-to-end ingestion, distributed GPU inference, and report generation with adaptive auto-tuning of batch size and worker concurrency enabling higher cost reduction.
- Demonstrated fault-tolerant high throughput scalability with zero errors across load tests; via Kafka-based backpressure and Spark auto-partitioning, enabling cost-effective assistance for radiologists in high-volume, resource-constrained clinical settings.